Computer Vision

(Classification, Object Detection, Segmentation)

What is computer vision?

- Computer vision enables machines to interpret, analyze, and pull meaningful data from images and videos. - ***Microsoft***

- Computer vision is a subfield of artificial intelligence (AI) that equips machines with the ability to process, analyze and interpret visual inputs such as images and videos. - ***IBM***

- **Using Computers to understand images.**

- Technically speaking, analysis is done with the help of algorithms many of which are AI-based.

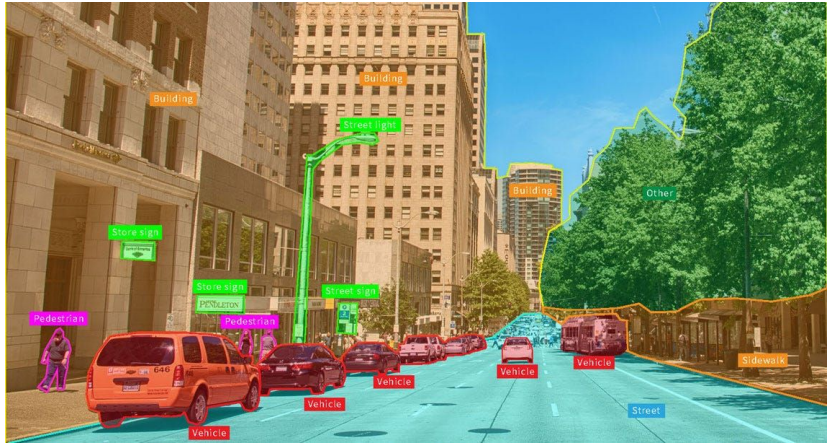
What can we do with Computer Vision

- There are many applications of computer vision such as;
  - Facial recognition
  - Disease diagnostics
  - Augmented Reality
  - Autonomous action (robotics)
  - Scene captioning

- Wide application in different domains;
  - Health
  - Agriculture
  - Security
  - Spatial Analysis

- In many cases the goal of using computer vision is to replicate the human visual system with similar or better performance.

- This has helped improve the efficiency of various tasks which would not be possible if we relied on human capability alone.



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What is an Image

- a digital representation of a visual scene, structured as a 2D or 3D numerical matrix (array)
- Makes it possible for computation by computers.

Gray image

Part of the image matrix

52	80	95	103	116	131	141	145	139	138	145
42	67	75	91	100	117	142	148	145	146	148
65	114	103	129	145	145	141	132	143	154	149
81	119	107	141	150	141	141	139	165	155	146
96	117	118	155	147	138	157	151	157	152	142
92	82	140	158	139	142	152	154	162	141	133
116	77	123	136	127	143	156	150	149	139	131
120	78	116	126	122	144	159	148	139	134	132
123	90	126	135	126	143	161	152	138	130	134
144	100	131	137	129	146	168	158	137	130	134
146	102	120	126	130	153	171	156	130	130	131

Types of Computer Vision Tasks

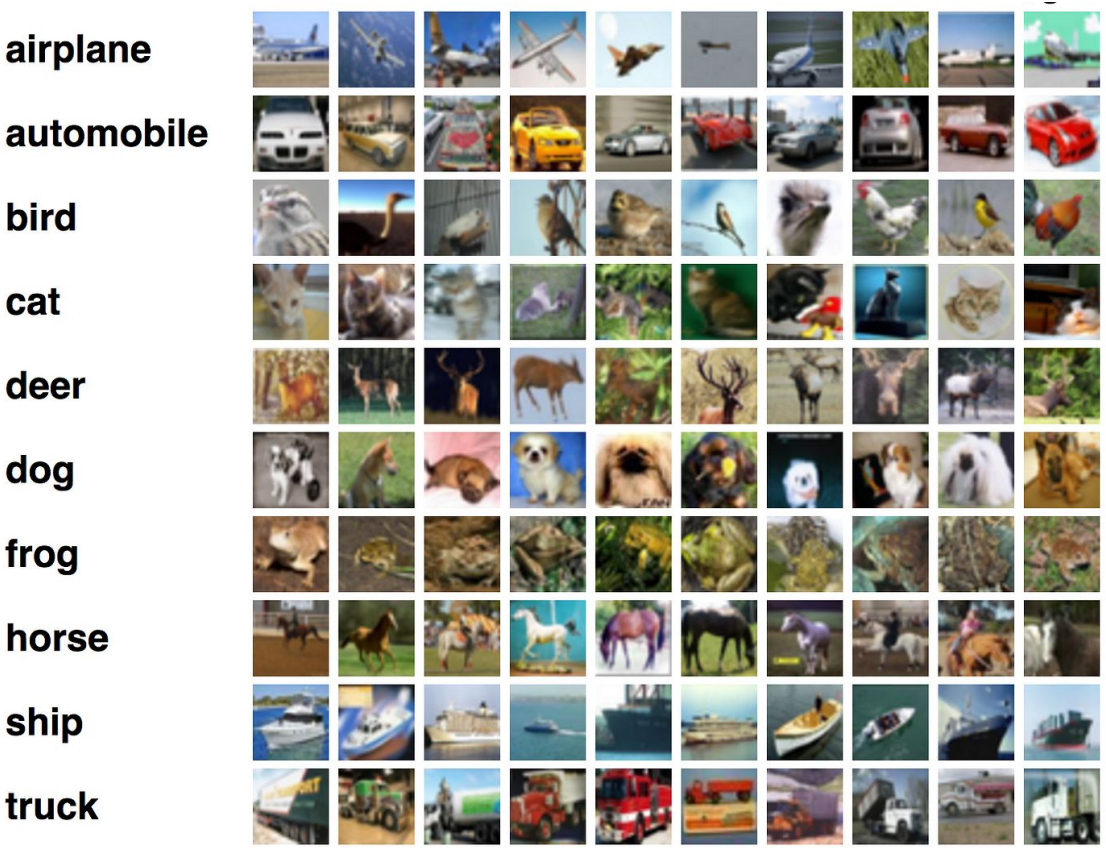
- Three common tasks
  - Image Classification
  - Object Detection/Localization
  - Image Segmentation

- Other tasks include;
  - Pose estimation
  - Image captioning,
  - Image tracking
  - Image enhancement

Image Classification

- Assign a predefined label to an entire image.

- The CV algorithm analyzes the image and learns features that represent a given class.



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Object Detection

- Beyond classification, it involves localization of objects
- Can identify multiple objects in the same image.
- Important when positional information is important e.g classifying multiple objects in an image

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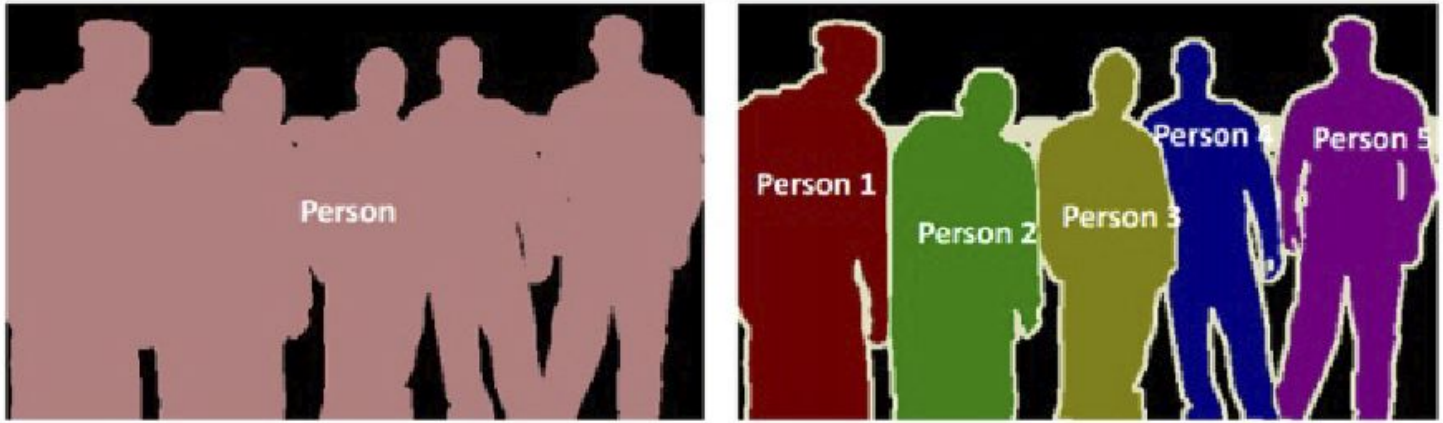
Image Segmentation

- Classification/partitioning at a pixel level
  - Semantic segmentation - classifying each pixel into a class.
  - Instance segmentation - distinguishing between pixels of instances/objects of the same type.



Semantic Segmentation Instance Segmentation

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Approaches to Computer Vision

- A bit of history:
  - 1950 - early theory, AI foundations for the neural network
  - 1970 - algorithms for feature detection e.g edge detection (canny edge, hough transformations)
  - 1980 - object detection algorithms - cascades, Scale Invariant Feature Transform (SIFT), clustering algorithms (gaussian mixture model)
  - 2000 - Support Vector machines, Boosting
  - 2000 - the great breakthrough
    - Large datasets and computational power
    - Convolutional Neural Networks (CNNs) e.g AlexNet
    - New techniques (augmentation, Generative Adversarial Networks)
    - Transformers (Attention is all you need) - Vision transformers, Visual LLMs
- Traditional
  - Detection - haar cascades, blob detection
  - Segmentation - Thresholding, watershed,
  - Classification - Colour histograms
- Modern

Computer Vision Algorithms (Traditional vs Modern)

- The main differentiator is based on the feature extraction.
- Traditional algorithms rely on manual feature extraction while the modern ones use automated approaches.
- Traditional algorithms include; haar cascades, SIFT, SVMs, Clustering,..
- Modern algorithms include: Deep Learning (CNNs, Transformers)

Computer Vision Algorithms

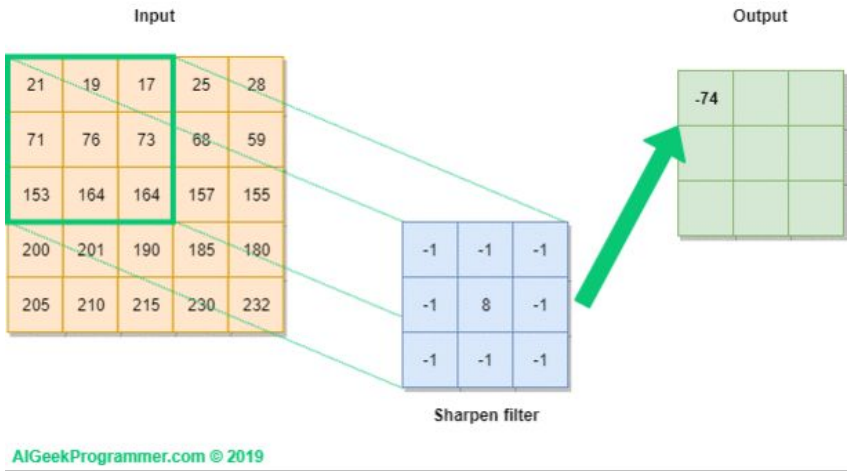
- Traditional vs Modern approaches
- Consider the following aspects
  - Data availability
  - Computational Requirements
  - Feature extraction/understanding
  - Interpretability
  - Performance (Accuracy)
  - Deployability

Computer Vision Algorithms

Factors	Traditional	Modern
Data Requirement	Low	High
Compute	Low	High
Feature extraction	Manual(limited)	Automated(unbounded)
Interpretability	High	Low
Deployability	Low	High

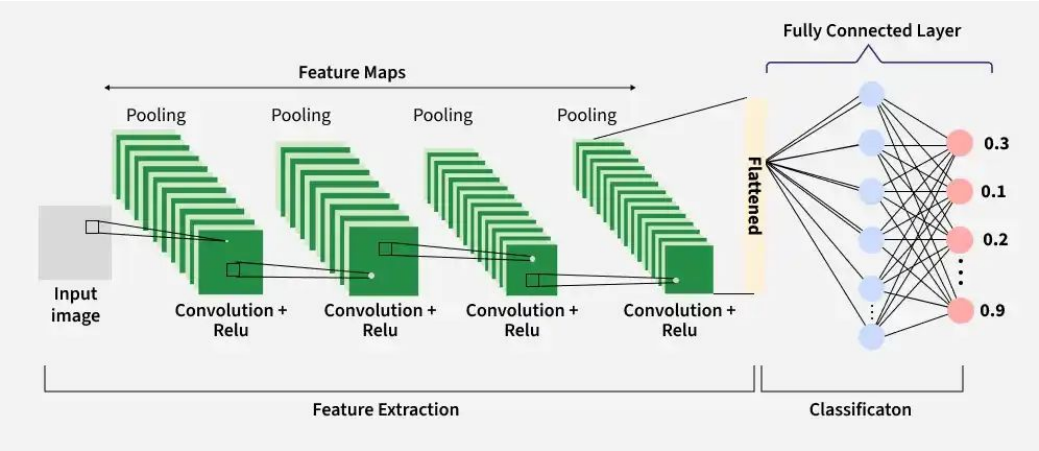
Convolutional Neural Networks

- These are deep learning models specialized for processing grid like data (e.g images)
- This processing is done using convolutions.
- Through this the algorithms learn complex representations of features



Convolutional Neural Networks

- The structure includes
  - Convolutional Layers - sliding kernels which generate feature maps
  - Activation Layers - learning units , enable non-linearity, learning complex features
  - Pooling Layers - pick most salient features, reduced computation
  - Fully connected layer - interpret extracted features for final tasks e.g. classification



Transformers

- Current SOTA for processing sequential data
- First popularized for Natural Language Processing (NLP)
- Key components include;
  - Self attention mechanism - capturing entire sequence of data and importance components.
  - Parallelization - analyze different inputs at the same time.
  - Encoder or Decoder - process input and output

CNN vs Transformer

- Analysis mechanism - CNN's use convolutions in sequence while Transformers analyze entire image at once as patches learning more complex relations.

- Data - Less data typically for CNNs vs Transformers

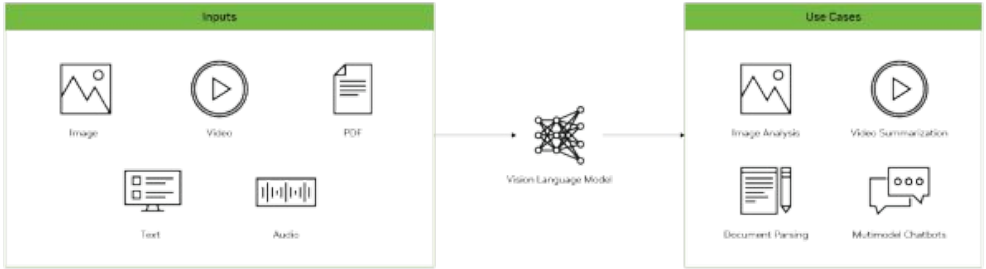
- Computational Efficiency - CNNs tend to require less resources

- Performance - a mixed bag but CNNs may be best for simple tasks while Transformers might perform better for complex tasks.

Vision Language Models (VLMs)

What are Vision Language Models

Vision language models (VLMs) are multimodal, generative AI models capable of understanding and processing video, image, and text.



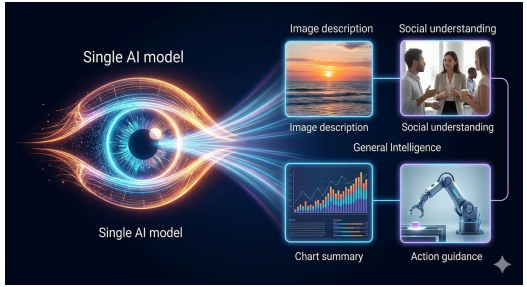
What Makes Up a Vision Language Model?

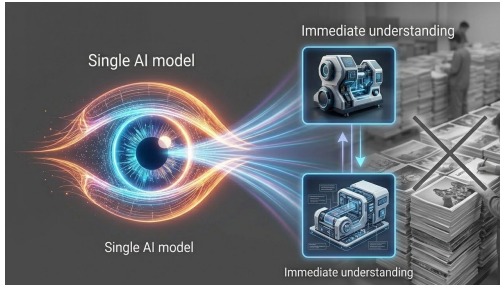
A **Vision Language Model (VLM)** is an AI system that integrates a **large language model (LLM)** with a **vision encoder**.

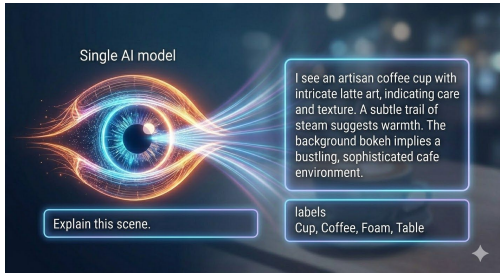
This combination allows the model to process visual data (images and videos) alongside text, enabling it to "see," interpret, and generate text-based insights based on those combined inputs.

Why Vision Language Models (VLMs) Important

- **One Tool for Many Jobs:** Old AI could only do one thing (like spot a cat). New VLMs are "all-rounders" that can describe, analyze, and answer questions about any image or video.
- **No More Expensive Training:** With old AI, you had to teach it every new thing by showing it thousands of examples. VLMs are already "smart" enough to understand new things instantly.
- **AI That Can "Talk":** Instead of just giving you a code or a label, a VLM can chat with you and explain what it sees, just like a person would.
- **Deeper Understanding:** It doesn’t just see shapes or read letters; it understands what those things actually *mean* in the real world.







Tailoring VLMs to Sector-Specific Needs

Health Data Collection:

- **Medical Imaging and Reports:** Compile vast datasets of medical images paired with diagnostic reports.
- **Patient Interaction Data:** Use for training models in understanding patient queries and health advice.

Agricultural Data Collection:

- **Crop, Soil, Pest Imagery:** Gather with farmer insights for context.
- **Local Knowledge Integration:** Engage local health workers and farmers for annotations.

Model Adaptation:

- **Fine-Tuning on Health and Agronomic Data:** Use techniques like PEFT for domain-specific tasks.
- **Prompt Engineering:** Design prompts for accurate medical diagnosis or crop management suggestions.

Challenges:

- Data privacy in health, lack of specific agricultural datasets, ensuring model accuracy and relevance.

You Only Look Once (YOLO)

- Is a Single Shot Detector
- First designed in 2015, it has undergone several improvements with over 8 versions published to date
- The biggest attribute is speed and accuracy.
- For many object detection tasks, it is an easy out of the box solution to finetune.
- Easy to deploy, especially for edge devices.

Real-World Applications

Health:

- **Medical Image Analysis:** Aid in diagnosing diseases from scans, reducing diagnostic time. Also, for image segmentation in tumor detection and organ localization.
- **Patient Education:** Translate complex medical information into patient-friendly visuals and texts.
- **Remote Monitoring:** Assist in interpreting patient data from home devices.

Agriculture:

- **Crop Health Monitoring:** Identify diseases or nutritional deficiencies in crops.
- **Precision Farming:** Offer advice on planting, irrigation, or pest control based on visual data.
- **Livestock Health:** Monitor animal health through image analysis for early disease detection.

Let’s go practical

- What is an Image
  - https://pixspy.com/
  - https://www.csfieldguide.org.nz/en/interactives/pixel-viewer/
- How a CNN works
  - https://poloclub.github.io/cnn-explainer/

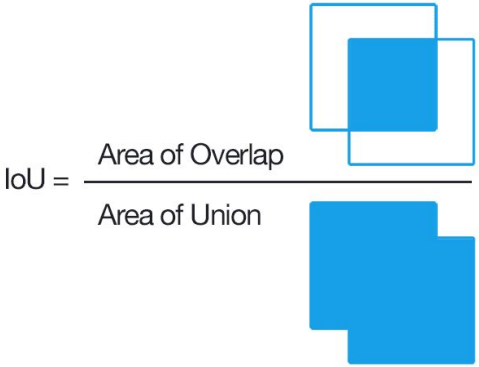
CNNs for Object Detection

- This is typically a dual stage approach which involves classification and bounding box generation.
- We have two main categories
  - Single Stage Detectors - bounding boxes are detected directly at the same time with class probabilities. e.g. You Only Look Once (YOLO), Single Shot Detector (SSD), RetinaNet
  - Two stage Detectors - First identify areas of interest in an image (anchors) for classification of presence of objects (usually done by a classification CNN) and then perform bounding box regression . e.g. R-CNN, Fast R-CNN, Faster R-CNN

CNNs for Object Objection - Evaluation

- Intersection Over Union (IOU) - measures overlap between ground truth box and the predicted box.

- Mean Average Precision (mAP) - a standard and good reflection of overall model accuracy, measures precision for each class.
  Typically represented across IOU thresholds e.g . a common measure for benchmarking is mAP@0.5 IOU.



You Only Look Once (YOLO)

- Four main stages
  a. Split the image into an SxS grid - also known as residual blocks, the image is split into equal patches in a grid
  b. Bounding box regression - for each of the patches/residual blocks, bounding boxes (fixed number) are predicted for objects in the block. It also predicts the probability classes of objects existing in each of the of the blocks.
     This results in an object having multiple bounding boxes.
  c. Box Refinement - predicted boxes with IOU above a set threshold are retained as these are the boxes that contain most of the object. **NOTE**: The IOU threshold is a hyperparameter and is typically set at 0.5
  d. Non Max Suppression - We may still have duplicate bounding boxes as there might be more than one box per object that met the IOU threshold. NMS is used to retain only boxes with the highest detection probabilities. **NOTE:** This can affect performance in scenarios where there are multiple clustered objects.

Resources

- YOLO paper
- https://medium.com/fruitpunchai/flying-high-with-ai-counting-pelican-breeding-pairs-in-the-danube-delta-ffb54cf99d8c
- https://github.com/roboflow/notebooks
- https://monai.org.cn/model-zoo.html
- https://supervision.roboflow.com/latest/